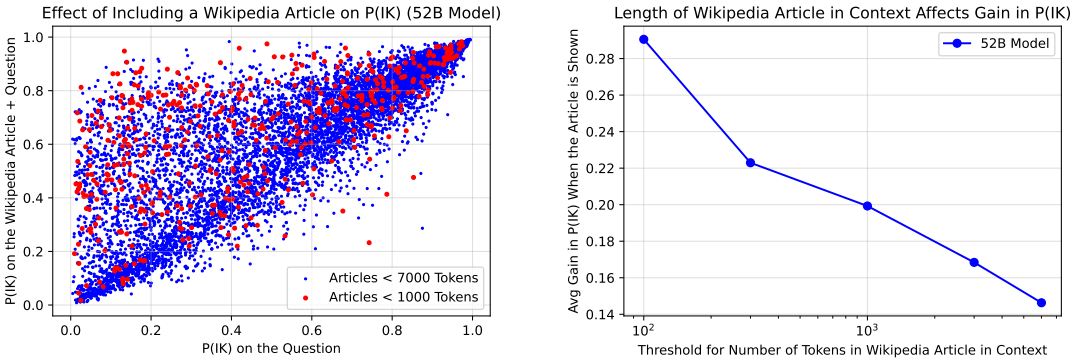

Figure 18 Effect of including Wikipedia article on P(IK) for TriviaQA Questions. We see that including a relevant Wikipedia article in the context boosts the average P(IK) on TriviaQA Questions. P(IK) increases more for shorter Wikipedia articles, from which it is presumably easier to identify the relevant facts.

then the P(IK) score rises to 78%. In other words, without any further training, P(IK) generalizes to address whether the language model can find answers to questions in source materials within its context.

We demonstrate this phenomenon quantitatively using questions from TriviaQA, by comparing P(IK) evaluated both with and without accompanying reference material. Questions from TriviaQA come with Wikipedia documents that are relevant to each questions. For each question, we then compute P(IK) both with an without the reference document. Specifically, the format for the prompts are as follows:

With Background Material:

```
Here is some background information material: <BACKGROUND_MATERIAL>
Now, answer the following question.
Question: Which Lloyd Webber musical premiered in the US on 10th
   December 1993?

Answer:
```

Without Background Material:

```
Question: Which Lloyd Webber musical premiered in the US on 10th
   December 1993?

Answer:
```

Whenever the full article was excessively long, we needed to truncate it to 7000 tokens so that the article and the question would always fit in the model context. Figure 18 summarizes the effects of including Wikipedia articles on P(IK). We see that including the article increases P(IK). Furthermore, shorter articles increase P(IK) more. Presumably this is because the correct answer is easier to extract from shorter articles than longer articles.

5.4 P(IK) Generalizes to Account for Hints Towards GSM8k Solutions

In this section we study how hints towards the solution of GSM8k problems affect P(IK) scores. Specifically, we add hints to the problem statement using the following format:

```
Question: Students in class 3B are collecting school points for behavior
   . If they get enough points in total, they can go on a trip. In the
   class, there are Adam, Martha, Betty, and Tom. Adam has collected 50
    points. Betty was better than Adam and collected 30% more. Marta
   managed to collect 3 times more points than Tom, who has 30 points
   less than Betty. How many points is the class missing to go on the
   trip if the minimum threshold is 400 points?
```

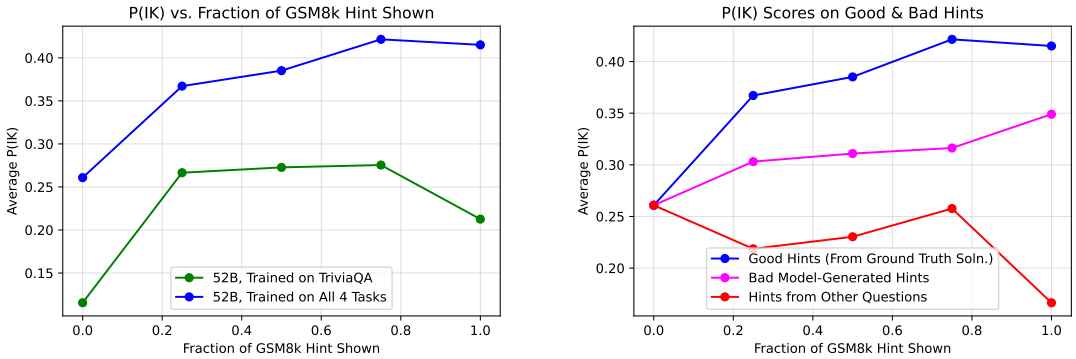

Figure 19 Effect of hints on P(IK) applied to GSM8k – all of these results represent generalization, as the models were not trained on GSM8k. Left: We see that showing more of the GSM8k hint results in higher P(IK). The effect is more consistent for the model trained on all 4 tasks (TriviaQA, LAMBADA, Arithmetic, and Python Function Synthesis), rather than the one trained on just TriviaQA. Right: We evaluate the model trained on all 4 other tasks on various hints. We see lower P(IK) scores for bad hints (though the models are partially fooled), and actual decreases in the P(IK) score when the hints are irrelevant because they come from other questions.

```
Here is a hint: Betty has 30% more points than Adam, so it's 30/100 * 50
    = <<30/100*50=15>>15 points more.
Betty's total is therefore 50 + 15 = <<50+15=65>>65 points.
Tom has 30 points less than Betty, so he has 65 - 30 = <<65-30=35>>35
   points.
Marta has 3 times more points than Tom, so she has 3 * 35 =
   <<3*35=105>>105 points.
In total, all students collected 50 + 65 + 35 + 105 =
   <<50+65+35+105=255>>255 points.
So the class is missing 400 - 255 = <<400-255=145>>145 points to go on
   the trip.

Answer:
```

We test two variations on hints. First, we vary the amount of information given away by the hint by truncating hints at fixed fractions of their total length. Second, we compare how P(IK) is affected by good, incorrect, and distracting hints. Bad hints are acquired by generating chain-of-thought samples from our 52B model and selecting only the samples that lead to an incorrect answer. Distracting hints are simply hints that were generated for a different question, so that they are irrelevant to the question at hand.

Figure 19 shows the result of evaluating GSM8k problems with hints using our 52B P(IK) model. We see that (1) showing more of the hint generally leads to higher P(IK), (2) good hints that lead to the correct answer result in higher P(IK) scores than bad hints, and (3) the P(IK) model that was trained on TriviaQA, LAMBADA, Arithmetic, and Python Function Synthesis performs better and more consistently, especially with partial hints.

5.5 Comparing Models Trained with Distinct Pretraining Distributions

We would like P(IK) to truly capture model self-knowledge. Thus we would like to distinguish between this and an alternative hypothesis, that P(IK) is merely capturing something like the intrinsic difficulty of tasks. In order to try to disentangle these explanations, we studied two 12B models (with identical architecture) that were pretrained on distinct data distributions – one was trained with four repetitions of a high quality dataset, while the other uses a single copy of that dataset mixed with a single copy of a larger but lower-quality distribution of webdata. We refer to these pretrained language models as A and B in this section. We finetuned both models for P(IK) on the questions from the TriviaQA training set that each model got correct.

These models get many of the same questions correct and incorrect. So our focus here will be on *the subset of questions that one model answers correctly and the other incorrectly*. In the TriviaQA test set there were

	Questions that only A gets right	Questions only B gets right
A's average P(IK)	0.463	0.408
B's average P(IK)	0.409	0.477

Table 2 'Cross-Experiments': Average P(IK) from for two distinct models on subsets of TriviaQA questions where one model is correct and the other is wrong. We see that the entries in the major diagonal are larger than the entries in the minor diagonal, showing that there is some signal that P(IK) is encoding model-specific information. However, the difference in P(IK) is only around 6%, so there is room for improvement from future work.

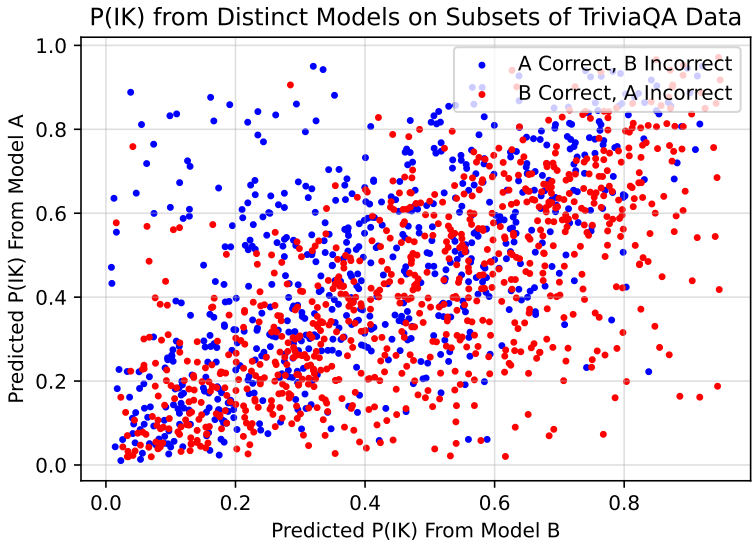

Figure 20 Scatterplot that disambiguates numbers in Table 2. We evaluate 2 distinct 12B models on TriviaQA, separating the data into questions each model gets correct and the other gets incorrect. The scatterplot depicts the P(IK) scores from each model on both of these data subsets.

650 and 826 such questions, respectively, that model A or B got correct but that the other got wrong. Table 2 shows that each model has a noticeably higher P(IK) for the questions that they get correct, compared to the questions that the other model gets right. In Figure 20 we show a scatter plot of P(IK) for models A and B, colored by which model got the questions right. The differences between the distributions are small but noticeable.

We also tried a second experiment, in order to try to get insight as to whether P(IK) leverages features specifically relevant to self-knowledge. For this purpose we finetuned *both* the A and B pretrained models on the ground-truth P(IK) data *from each model*. So for example, we started from model A, and finetuned it on the ground-truth P(IK) data from model A and, separately, on the data from model B (resulting in two distinct finetuned snapshots). We might hope to find that training a P(IK) classifier to predict whether or not model A knows the answer to a question would work better when starting from model A itself as the initial checkpoint, as compared to if we start from model B. Table 3 contains results for this experiment.

We find mixed but encouraging results here: when testing on the ground-truth P(IK) data from model A, both starting from model A and starting from model B seems to give comparable performance. However, when testing on ground-truth P(IK) data from model B, starting from model B seems to do better than starting from model A. We also tried focusing on only the questions where the two models differ, but we found similar results in that case.